MSE 315: Introduction to Computational Materials Engineering

Sem-II AY 2025-26

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Class Information and Meeting Times

Lectures (7/210): G1 (11:30 – 12:50 PM, Tuesday) and **G2** (11:30 – 12:50 PM, Friday)

Labs/Tutorials (7/202): I1 (2:00 PM – 3:20 PM, Monday) and **I2** (2:00 PM – 3:20 PM, Wednesday)

Objectives:

- Appreciate and understand the complexity of materials including materials design and processing, via the lens of **computational models and simulations**.
- Learn **fundamental theory** governing various computational techniques and apply these to simulate materials and **solve Materials Science related and interdisciplinary problems**.
- Run **state-of-the-art simulations across various length and time scales**.
- Develop a feel for reading and understanding various **relevant and interesting computational materials publications** and incorporate some of them via hands-on exercises.

Course Contents:

- Introduction: Role of computational techniques; Materials design and selection and computational databases
- A simplistic materials model: the Random-walk diffusion model
- Ab-initio methods and electronic structure *
- Molecular Simulations
  - Molecular dynamics*
  - Monte Carlo methods*
- Phase Field methods*
- Continuum Methods: Finite difference and Finite Element Methods*
- Materials Informatics and data-driven approaches

Reference Material:

Main Text: *Introduction to Computational Materials Science: Fundamentals to Applications* - Richard LeSar

Supplemental Reference books:

1. *Computational Materials System Design* by Dongwon Shin and James Saal

2. *Computational Materials Discovery* by Artem R Oganov, Gabriele Saleh and Alexander G Kvashnin

Assessment:

Assignments (25%) – 5 assignments with weightage of 5% each

Exam I (25%)

Surprise quizzes (15%)

Term Project (25%)

Class participation and attendance (10%) – Attendance is mandatory unless there is a justified medical reason for missing a class supported by a doctor's certificate.

Tutorial:

Tutorial will aid in homework, additional demos and running/debugging simulations and clearing conceptual doubts. Attendance in tutorials is mandatory.

Lab:

Lab will involve perfuming and analyzing Materials simulations using a variety of software and tools. Attendance in labs is mandatory.

Lecture Materials and Assignments:

Google Classroom will be used for sharing lecture materials and readings. Assignments will be uploaded and submitted via this portal.

Honor code and collaboration:

As you are aware, the institute has zero tolerance towards academic dishonesty of all kinds including cheating, seeking or providing unauthorized help on any evaluated work and plagiarism. Please adhere to the student honor code during all aspects of this course. Doing the homework assignments in this class is essential to mastering the concepts. It is ok to discuss homework assignments with others for purposes of developing a better understanding, but the **work turned in should be your own.**